

Definition C.8 (Counting Function). For index set $I_1, I_2 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$ and index i , a counting function $\text{COUNT}[I_1, I_2, i]$ satisfies the following, if $\forall v \in I_1 \cup I_2, k \in [l], X_{v,k} \in \mathbb{Z}_p$ and $X_{v,k} \neq 0$, then

$$\begin{aligned} \text{COUNT}[I_1, I_2, i](X)_{i,k} &= \frac{1}{\sum_{h=1}^k \mathbf{1}[X_{I_1,h} = X_{I_2,k}] + 1} \quad \forall k \in [l]. \\ \text{COUNT}[I_1, I_2, i](X)_{i^c,k} &= 0 \quad \forall k \in [l]. \end{aligned}$$

Definition C.9 (Matching Function). For index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$, a matching function $\text{Match}[I_1, I_2, I_3, I_4]$ satisfies the following, if $\forall v \in I_1 \cup I_2, k \in [l], X_{v,k} \in \mathbb{Z}_p$, then

$$\begin{aligned} \text{Match}[I_1, I_2, I_3, I_4](x)_{I_3,k} &= X_{I_4,k^*} \quad \forall k \in [l] \\ \text{where } k^* &= \begin{cases} \min\{h \mid X_{I_1,h} = X_{I_2,k}\}, \{h \mid X_{I_1,h} = X_{I_2,k}\} \neq \emptyset \\ 1, \text{ otherwise} \end{cases} \end{aligned}$$

Definition C.10 (Matching Closest Function). For index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$, a matching closest function $\text{Match}[I_1, I_2, I_3, I_4]$ satisfies the following, if $\forall v \in I_1 \cup I_2, k \in [l], X_{v,k} \in \mathbb{Z}_p$, then

$$\begin{aligned} \text{MatchClose}[I_1, I_2, I_3, I_4](x)_{I_3,k} &= X_{I_4,k^*} \quad \forall k \in [l] \\ \text{where } k^* &= \begin{cases} \max\{h \mid X_{I_1,h} = X_{I_2,k}\}, \{h \mid X_{I_1,h} = X_{I_2,k}\} \neq \emptyset \\ 1, \text{ otherwise} \end{cases} \end{aligned}$$

Definition C.11 (Matching Nearest Function). For index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$ and index i , a matching nearest function $\text{MatchNearest}[I_1, I_2, I_3, I_4, i]$ satisfies the following, if $\forall v \in I_1 \cup I_2, k \in [l], X_{v,k} \in \mathbb{Z}_p$, then

$$\begin{aligned} \text{MatchNearest}[I_1, I_2, I_3, I_4](x)_{I_3,k} &= X_{I_4,k^*} \quad \forall k \in [l] \\ \text{where } k^* &= \begin{cases} \arg \min_{h \in \{h \mid X_{I_1,h} = X_{I_2,k}\}} |h - X_{i,k}|, \{h \mid X_{I_1,h} = X_{I_2,k}\} \neq \emptyset \\ 1, \text{ otherwise} \end{cases} \end{aligned}$$

Definition C.12 (Matching Next Function). Given any interger constant A , assuming $p > 10A \log n$, for index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$, and a special counting index a , a matching next function $\text{MatchNext}[I_1, I_2, I_3, I_4, a]$ satisfies the following, if X satisfies the following condition:

1. $\forall v \in I_1 \cup I_2, k \in [l], X_{v,k} \in \mathbb{Z}_p$,
2. $X_{a,k} \in \text{ROUND}(1/[n^A], p) \cup \{0\}$,
3. For any $k \in [l]$, given any $k \geq k$, the counting index multiset $S_k = \{X_{a,k'} \mid X_{I_1,k'} = X_{I_2,k}\}$ takes consecutive and disjoint values in $\text{ROUND}(1/[n^A], p)$, that is, there exists $u_k, v_k \in \text{ROUND}(1/[n^A], p)$ such that $S_k = [u_k, v_k] \cap \text{ROUND}(1/[n^A], p)$.

then, we have

$$\begin{aligned} \text{MatchNext}[I_1, I_2, I_3, I_4, a](X)_{I_3,k} &= X_{I_4,k^*} \quad \forall k \in [l] \\ \text{where } k^* &= \arg \min_{h \in \{h \mid X_{I_1,h} = X_{I_2,k}\} \cup \{1\}} |X_{a,h} - \text{next}(X_{a,k})|. \end{aligned}$$

Now we will show how to implement these functions using Transformer Blocks. The construction here is motivated by the construction in [Feng et al. \(2023\)](#) with some modifications.

Lemma C.13 (Copying Blocks). For integer s , index set $I_1, I_2 \subset [d - 10]$ satisfying $|I_1| = |I_2|$, a copying function $\text{COPY}[s, I_1, I_2]$ can be implemented with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k$, it holds that

$$\mathcal{A}(g(X) + X) = \text{COPY}[s, I_1, I_2](X).$$

Proof of Lemma C.13. We will use the feedforward block to calculate $X_{k,d}^2$ and 1 (Lemma C.1) and have

$$\begin{aligned} (g(X) + X)_{d-1,k} &= k^2 \\ (g(X) + X)_{d-2,k} &= 1. \\ \forall i \notin \{d-1, d-2\}, (g(X) + X)_{i,k} &= X_{i,k}. \end{aligned}$$

We will use X' to denote $g(X) + X$. Then we will choose $W^{(K)}, W^{(Q)}$ such that

$$W^{(K)} X'_{:,k'} = n \begin{bmatrix} 1 \\ k' \\ k'^2 \end{bmatrix}$$

$$W^{(Q)} X'_{:,k} = \begin{bmatrix} -(k^2 + s^2 - 2sk) \\ 2k - 2s \\ -1 \end{bmatrix}$$

Hence

$$\begin{aligned} & \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} \\ &= -n (k'^2 - k'(2k - 2s) + k^2 + s^2 - 2sk) \\ &= -n(k - s - k')^2 \end{aligned}$$

Hence we have

$$\arg \max_{k' < k} \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} = \max\{k - s, 0\}.$$

Also, for any $k' \leq k, k' \neq \max\{k - s, 0\}$, we have

$$\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} - \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{\max\{k-s,0\},k} < -n.$$

Hence after the column-wise softmax and rounding to $p = O(\log n)$ bit, we have

$$\left(\text{softmax} \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right) \right)_{k',k} = \mathbf{1}[k' = \max\{k - s, 0\}].$$

We will then choose $W^{(V)}$ such that

$$\begin{aligned} W^{(V)} X'_{I_2,k'} &= X'_{I_1,k'} = X_{I_1,k'} \quad \forall k' \in [l]. \\ W^{(V)} X'_{I_2^c,k'} &= 0 \quad \forall k' \in [l]. \end{aligned}$$

This then concludes that

$$\mathcal{A}(g(X) + X) = \text{COPY}[s, I_1, I_2](X).$$

The proof is complete. $\square$

Lemma C.14 (Counting Blocks). *For index set $I \subset [d - 20]$ satisfying $|I_1| = |I_2| \leq 10$, a counting function $\text{COUNT}[i, I_1, I_2]$ can be approximated with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k$ and $X_{3,k} = \mathbf{1}[k = 1], X_{I_1,1} = 0$, it holds that*

$$\begin{aligned} \mathcal{A}(g(X) + X)_{i,k} &= \text{ROUND}(\text{COUNT}[s, I_1, I_2](X)_{i,k}, p). \\ \mathcal{A}(g(X) + X)_{i^c,k} &= 0. \end{aligned}$$

Proof of Lemma C.14. We will use the feedforward block to calculate $X_{v,k}^2, v \in I_1 \cup I_2$ (Lemma C.1) and have

$$\begin{aligned} (g(X) + X)_{d-i,k} &= X_{I_1[i],k}^2, i \in [|I|]. \\ (g(X) + X)_{d-|I|-i,k} &= X_{I_2[i],k}^2, i \in [|I|]. \\ (g(X) + X)_{d-2|I|-1,k} &= 1. \\ \forall i \notin \{d - i \mid i \in [2|I| + 1]\}, (g(X) + X)_{i,k} &= X_{i,k}. \end{aligned}$$

We will use X' to denote $g(X) + X$. Then we will choose $W^{(K)}, W^{(Q)}$ such that

$$W^{(K)} X'_{:,k'} = n \begin{bmatrix} 1 + \mathbf{1}[k' = 1] \\ X_{I_1[i],k'} \\ X_{I_1[i],k'}^2 \end{bmatrix}_{i \in [I]}$$

$$W^{(Q)} X'_{:,k} = \begin{bmatrix} X_{I_2[i],k}^2 \\ -X_{I_2[i],k} \\ 1 \end{bmatrix}_{i \in [I]}$$

Hence,

$$\begin{aligned} & \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} \\ &= -n \sum_{i=1}^{|I|} \left(X_{I_2[i],k'}'^2 - X_{I_1[i],k'} (2X_{I_2[i],k}) + X_{I_2[i],k}^2 \right) + n \mathbf{1}[k' = 1] \sum_{i=1}^{|I|} X_{I_1[i],k}^2 \\ &= -n \sum_{i=1}^{|I|} (X_{I_1[i],k'} - X_{I_2[i],k})^2 + n \mathbf{1}[k' = 1] \sum_{i=1}^{|I|} X_{I_2[i],k}^2. \end{aligned}$$

Hence we have

$$\max_{k' < k} \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} = 0.$$

Equality holds when $k' = 1$ or $X_{I_1[i],k'} = X_{I_2[i],k}$ for all $i \in [|I_1|]$.

Also, for any $k' \leq k, k' \neq 1$ or $X_{I_1[i],k'} \neq X_{I_2[i],k}$ for some $i \in [|I_1|]$, we have

$$\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} < -n.$$

Hence after the column-wise softmax and rounding to $p = O(\log n)$ bit, we have

$$\left(\text{softmax} \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right) \right)_{k',k} = \text{ROUND} \left(\frac{1}{\sum_{h=1}^k \mathbf{1}[X_{I_1,h} = X_{I_2,k}] + 1}, p \right)$$

Here the $O\left(\frac{1}{n^\alpha}\right)$ term comes from the fact that the softmax is rounded to $p = O(\log n)$ bit.

We will then choose $W^{(V)}$ such that

$$W^{(V)} X'_{i,k'} = X'_{3,k'} = \mathbf{1}[k' = 1] \quad \forall k' \in [l].$$

$$W^{(V)} X'_{I^c,k'} = 0 \quad \forall k' \in [l].$$

This then concludes that

$$\mathcal{A}(g(X) + X)_{i,k} = \text{ROUND}(\text{COUNT}[s, I_1, I_2](X)_{i,k}, p).$$

$$\mathcal{A}(g(X) + X)_{i^c,k} = 0.$$

□

Lemma C.15 (Matching Blocks). *Given any constant c , for index set $I_1, I_2, I_3, I_4 \subset [d - 20], |I_1| = |I_2| \leq 10, |I_3| = |I_4|$, a matching function $\text{Match}[I_1, I_2, I_3, I_4]$ can be implemented with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k, X_{3,k} = \mathbf{1}[k = 1], X_{I_1,1} = 0$ and $k \leq n^c$, it holds that*

$$\mathcal{A}(g(X) + X) = \text{Match}[I_1, I_2, I_3, I_4](X)$$